

Assessing Sea lice infection levels in Irish Atlantic salmon farms: Metrics and treatment trigger levels

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ABSTRACT

Sea lice, ectoparasites of Atlantic salmon (*Salmo salar*), are considered one of the main limiting factors for Atlantic salmon aquaculture. To ensure low infection levels and effective treatments, a wide range of management strategies are in place, integral to which is a monitoring programme. Treatment trigger levels set within each monitoring programme represent the threshold in sea lice infection that triggers a treatment to reduce and maintain low sea lice abundance levels. This study investigated sea lice infection metrics on Atlantic salmon farms in Ireland and the effect of treatment trigger levels on the accuracy of reported sea lice counts. A linear relationship between sea lice abundance and prevalence was observed up to prevalence levels of ~40 %. Although using prevalence for reporting would be beneficial in cases of low infection levels and as a resource-efficient approach, reporting both abundance and prevalence is more informative. Reporting and targeting total adult female sea lice instead of ovigerous females for treatment trigger levels in Ireland would lead to better comparability of infection levels between countries. Although previous studies described under-reporting in the monitoring performed by the industry's producers, this bias has not been observed in the sea lice abundance data in Ireland. Lastly, no discrepancies of reported counts around treatment trigger levels were detected indicating that treatment trigger levels have no negative impact on the accuracy of reporting. Accurate estimation of sea lice population levels on salmon farms is needed for effective management including triggering sea lice treatments.

1. Introduction

Sea lice, *Lepeophtheirus salmonis*, are considered one of the most limiting factors of Atlantic salmon aquaculture due to the cost of management strategies (Liu and Bjelland, 2014; Torrisen et al., 2013). Management strategies aim to maintain low infection levels and ensure effective treatments (Overton et al., 2019; Boerlage et al., 2024), while also protecting fish welfare and wild salmon populations (Ives et al., 2023; Vormedal, 2024). Management strategies encompass integrated pest management, which includes veterinary medicinal and non-medicinal treatments—such as preventive methods, physical removal, and biological control—alongside monitoring programs (Cerbule and Godfroid, 2020; Erkinharju et al., 2021; Guragain et al., 2021; Oldham et al., 2023; Guttu et al., 2024; Thompson et al., 2024). Monitoring of sea lice abundance is an integral part of all management strategies

(Treasurer and Pope, 2000; Jeong et al., 2023) and infection levels established during monitoring are used to ensure effective timing of treatments (Westcott et al., 2004). The monitoring programme is either carried out by the individual salmon producers under self-reporting or by an independent third party (Klimesova et al., 2025; Rabe et al., 2024; Vormedal, 2024). In Ireland the Marine Institute (MI) has operated the National Sea lice Monitoring Programme since 1991 (NSLMP) (Jackson et al., 1997; DAFM, 2008; D'Arcy et al., 2024).

To ensure that sea lice abundance is kept at low levels, a threshold value has been established in most jurisdictions where marine Atlantic salmon farming takes place. This value represents the level of sea lice infection which should not be exceeded and, in instances where it is exceeded, it triggers an action which should be taken in order to reduce infection levels on a particular farm (Coates et al., 2021; D'Arcy et al., 2024). Different salmon producing countries implement different trigger

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levels, with many including stricter trigger levels during the wild salmonid smolt outmigration period (D'Arcy et al., 2024; Jeong et al., 2023; Vormedal, 2024). In Ireland for example, the treatment trigger level is set to 0.5 ovigerous females per fish at the farm level in spring during the outmigration and at two ovigerous females per fish at the farm level for the rest of the year (D'Arcy et al., 2024). Many industry producers perform their own monitoring independent of the NSLMP. For example, Mowi Ireland uses a different sampling strategy compared to the MI (weekly sea lice monitoring compared to monthly/fortnightly), and their treatment trigger levels are set to a mean value of 0.2 total adult females per fish at the farm level for the whole year (MOWI Ireland, 2022).

The most common method to quantify sea lice infection levels is abundance (mean count of sea lice per fish at the farm level), although the use of prevalence (proportion of infected fish) has been discussed in literature (Bush et al., 1997; Thorvaldsen et al., 2019; D'Arcy et al., 2024; Jeong et al., 2023). Abundance is considered more informative, especially in cases of high infection levels, when prevalence often reaches 100 % (Todd et al., 2006; Baillie et al., 2009). However, using prevalence for reporting has its advantages. One of them is the resource efficiency of this approach, as only the absence or presence of sea lice is recorded. The reduced inspection time also implies a potential benefit for fish welfare, since less handling is required to determine prevalence (Baillie et al., 2009). Furthermore, an almost linear relationship between abundance and prevalence has been reported (Baillie et al., 2009; Heuch et al., 2011; Patanasatienkul et al., 2013), particularly in cases of low abundance levels. Therefore, it is possible to report only prevalence and still be able to estimate abundance levels (Heuch et al., 2011).

Even though the countries are united in their use of abundance for estimation of sea lice infection levels, sampling strategies differ among jurisdictions. Most salmon-producing regions follow the principle outlined by Revie et al. (2005), in which a small number of fish are sampled from multiple pens to account for inter-pen variability. In contrast, Ireland's NSLMP predates this framework and employs a consistent approach based on inspecting a designated reference pen together with one randomly selected pen at each site. This consistency facilitates long-term trend analyses but differs from the multiple pen-based approach adopted elsewhere (DAFM, 2008). In addition, the Irish monitoring programme focuses on the count of ovigerous females (as in Chile; DAFM, 2008; Sernapesca, 2022), in contrast to total adult females (Faroe Islands, Norway, Scotland; FAO, 2013; Scottish Government, 2020; ICES, 2023; Vormedal, 2024) or total mobiles (all adults and preadults; British Columbia (Canada); DFO, 2020).

Although most countries with Atlantic salmon production rely on sea lice abundance data reported by producers (Vormedal, 2024; Rabe et al., 2024), a bias connected with a low incentive for accurate reporting has been previously described in self-reported sea lice counts (Elmoslemany et al., 2013; Godwin et al., 2021). Several solutions were proposed to remove the bias from the self-reported sea lice monitoring such as an improvement in the quality of sampling (Westcott et al., 2004; Godwin et al., 2021), unannounced audits (Elmoslemany et al., 2013; Godwin et al., 2021) or monitoring by an independent third party. However, the effect of switching to an independent third party on the accuracy of sea lice monitoring has not been studied yet (Godwin et al., 2021).

In response to calls for stricter sea lice control measures, several countries have lowered their treatment trigger levels over the years (Gislason, 2018; Vollset et al., 2018). However, recent research has highlighted flaws in overly ambitious trigger levels. These can lead to excessive treatment, emergence of drug resistance (McEwan et al., 2016), while also negatively impacting fish welfare and increasing mortality rates (Overton et al., 2019). Additionally, lower thresholds may contribute to under-reporting of sea lice infections, as individuals tend to report values below the set limits (Jeong et al., 2023).

An accurate estimation of sea lice population levels on salmon farms is needed for effective management (Groner et al., 2016). In this study, sea lice abundance data collected independently as part of the NSLMP,

in addition to industry data collected by a salmon producing company (Mowi Ireland), were analysed to determine the suitability of using prevalence as a measure of sea lice infection levels by exploring the relationship between abundance and prevalence. To gain a better understanding of the relationship between ovigerous females and total adult females, the reported counts of both life stages were examined. The data were also used to compare the accuracy of sea lice counts reported under the NSLMP with the industry producer's data. Lastly, the influence of current treatment trigger levels on the accuracy of the reported sea lice counts was determined.

2. Methods

Sea lice abundance data used for this study were taken from two different sources: (1) independently collected data by the Marine Institute (MI) as part of the National Sea lice Monitoring Programme (NSLMP) and (2) self-reported industry producer's data provided by Mowi Ireland.

2.1. National sea lice monitoring programme (NSLMP)

Sea lice counts recorded as part of the NSLMP by the MI are collected once a month except for spring (March, April, May) and winter months (January, December). In spring, monitoring takes place twice a month and during the two winter months, only one inspection is carried out. During each inspection, 30 fish are sampled from each of two pens. One pen on each farm is dedicated as a standard which is sampled every time, and a second pen is chosen opportunistically at each inspection (DAFM, 2008; D'Arcy et al., 2024). Counts of all sea lice life stages are recorded, and all mobile life stages are collected into 70 % ethanol for confirmation of correct identification. In addition, the water with anaesthetic is sieved for detached sea lice which are also included in the counts. Counts of detached sea lice were excluded from the analysis for better comparison with the industry monitoring which does not include detached sea lice. Detached lice were recorded in 56 % of all inspected pens, and the overall mean proportion of detached to attached total mobile lice was 11 %.

The data used span the period from 2006 to 2023 and the counts of ovigerous females, total adult females and total mobiles recorded per individual fish examined were used. There are two gaps in the data due to the Covid-19 pandemic (2020: April to June; 2021: February to April), when the MI did not carry out the monitoring inspections on salmon farms. Outliers (7 counts per fish) were excluded from the analysis. The final dataset was collected from a total of 178,974 individual fish inspected for sea lice abundance from 39 Atlantic salmon farms, located along the west coast of Ireland (Fig. 1).

2.2. Industry producer's monitoring (Mowi Ireland)

Industry data provided by the producer Mowi Ireland, were collected from all active pens on a farm every week. The number of pens examined per inspection varied from one to 15, with an average of five. From each pen, 20 fish were visually examined, and sea lice abundance was recorded. For this study, industry data from 14 Atlantic salmon farms (Fig. 1) over the period October 2015 to January 2020 were used. These farms were also independently monitored under the NSLMP, meaning their sea lice counts appear in both datasets. Outliers (1 count per fish) were excluded from the analysis. The counts of total adult females and total mobiles (all adult and pre-adult life stages) were recorded from a total of 119,721 fish.

2.3. Data analysis

2.3.1. Prevalence as a metric of sea lice infection, relationship between ovigerous females and total adult females

Data from the NSLMP were examined to determine the relationship

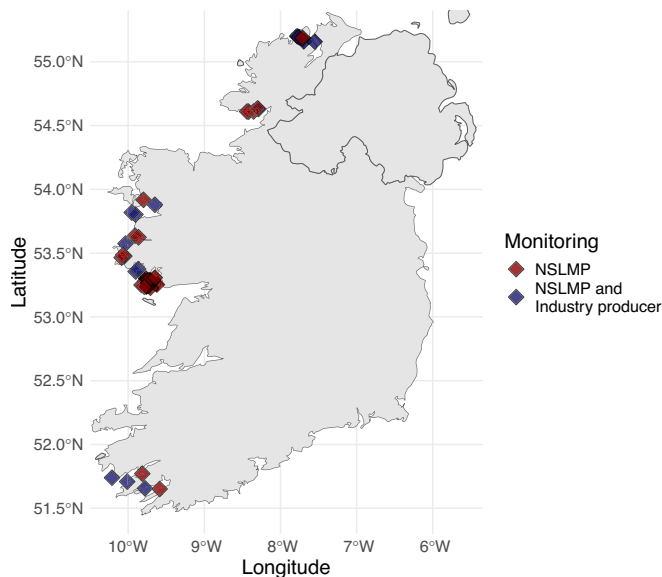


Fig. 1. Locations of Atlantic salmon farms monitored under the National Sea Lice Monitoring Programme (A) and by the industry producer (B), for this study.

between abundance and prevalence, and between ovigerous females and total adult females. Both analyses were conducted at the pen level. Abundance was calculated as a mean count of sea lice and prevalence as a mean of infected fish. Both metrics were calculated separately for total adult females and ovigerous females. Linear regression models were used to assess the relationship between prevalence and abundance for both ovigerous and total adult females. The index of dispersion (σ^2/μ) was also calculated separately for ovigerous and total adult female counts. To evaluate its association with abundance, dispersion index and abundance were computed for each year and salmon farm. Finally, a separate linear regression model was used to examine the relationship between ovigerous females and total adult females.

2.3.2. Differences in reported sea lice counts

Sea lice data from the NSLMP and industry producer monitoring were analysed to assess differences in reported sea lice counts. For both datasets, the mean abundance of total adult females and total mobiles per fish at the pen level was calculated. The NSLMP dataset was filtered to include only data from the same salmon farms and years as the industry producer's dataset (Oct 2015 – Dec 2019). To align the datasets, sea lice counts from individual pens in the NSLMP dataset were matched to the closest available inspection date in the industry producer's dataset. If multiple pens were sampled on the same farm and date, a pen different from the one inspected by the NSLMP was randomly selected. Sample points with inspection dates more than one week apart were excluded from the analysis (excluding 53 out of 662 data points).

Bland-Altman plots were used to visualise the difference between paired reported values by the NSLMP and industry producer's monitoring against their average (Bland and Altman, 1999). The mean difference was modeled as a function of the average, and the standard deviation of the differences was estimated from the absolute residuals of this regression. The 95 % limits of agreement were calculated as the predicted bias ± 2.46 times the predicted absolute residuals, providing limits that vary with the magnitude of the measurement. Non-parametric Wilcoxon matched pairs test (Wilbur, 1994) was used to compare the difference in medians between the two groups of reported sea lice counts.

2.3.3. Influence of treatment trigger levels

To determine the influence of treatment trigger levels on the accuracy of reporting, the sea lice data recorded as part of the NSLMP and industry monitoring were analysed. The NSLMP data were divided into two groups according to the time of the year with different treatment trigger levels (spring: 0.5 ovigerous females, summer to winter: two ovigerous females) and the mean abundance of sea lice (ovigerous females, total adult females) per fish at the pen level was calculated. In addition, the industry producer's counts of total adult females were compared to the NSLMP reported sea lice counts. To assure comparability of the two datasets, the NSLMP reported data were adjusted to contain only sea lice counts from the same salmon farms and months as the industry producer's data.

All statistical analyses were carried out using the statistical computing environment R 4.3.3 (R Core Team, 2024). The package 'ggplot2' for R was used for all data visualisation (Wickham, 2016) and 'rnatualearth' (Massicotte and South, 2025) was used for map visualisation.

3. Results

3.1. Prevalence as a metric of sea lice infection

The relationship between the measured values of abundance and prevalence for both ovigerous females and total adult females was initially linear, up to a mean abundance of approximately 0.5 and 0.6, respectively, corresponding to prevalence levels of ~40 % (Fig. 2). Beyond this point, the relationship became curvilinear, approaching an asymptote near 100 % prevalence at abundance levels of ~2 for ovigerous females and ~1.75 for total adult females. The index of dispersion was 6.4 for ovigerous females and 7.9 for total adult females, with dispersion increasing as abundance increased (Supplementary Information: Fig. 7). Both ovigerous and total adult females exhibited the same general trend, although prevalence rose more steeply at low abundance for ovigerous females, whereas total adult females showed a greater increase in prevalence at higher abundance levels.

3.2. Relationship between ovigerous females and total adult females

The mean abundance of ovigerous females and total adult females was assessed, and their relationship to treatment trigger levels was examined. The two variables were highly correlated (Fig. 3) with a correlation coefficient of 0.96. The equation of the fitted regression $OF = 0.71 TF - 0.02$ was used to calculate the corresponding treatment trigger levels between life stages. The corresponding treatment trigger levels for total adult females, based on the currently used thresholds for ovigerous females (a mean of 0.5 and 2 per fish at the pen level), were determined to be 0.7 for spring and 2.8 in the summer to winter. Conversely, if the existing trigger levels were applied to total adult females instead (0.5 and 2 per fish at the pen level), the equivalent mean values for ovigerous females would be 0.3 in spring and 1.4 in summer to winter.

3.3. A comparison of reported sea lice counts

Sea lice counts reported by the National Sea lice Monitoring Programme (NSLMP) and the industry producer were compared to determine whether any differences exist. Almost half of the pens sampled under the NSLMP and by the industry producer were sampled on the same date (45 %) and the mean of absolute difference between the dates of monitoring was 1.5 days after excluding outliers (sample points that were more than one week apart, $n = 15$). The correlation between the absolute difference of days between monitoring and the absolute differences in reported sea lice counts was -0.15 with 95 % Confidential

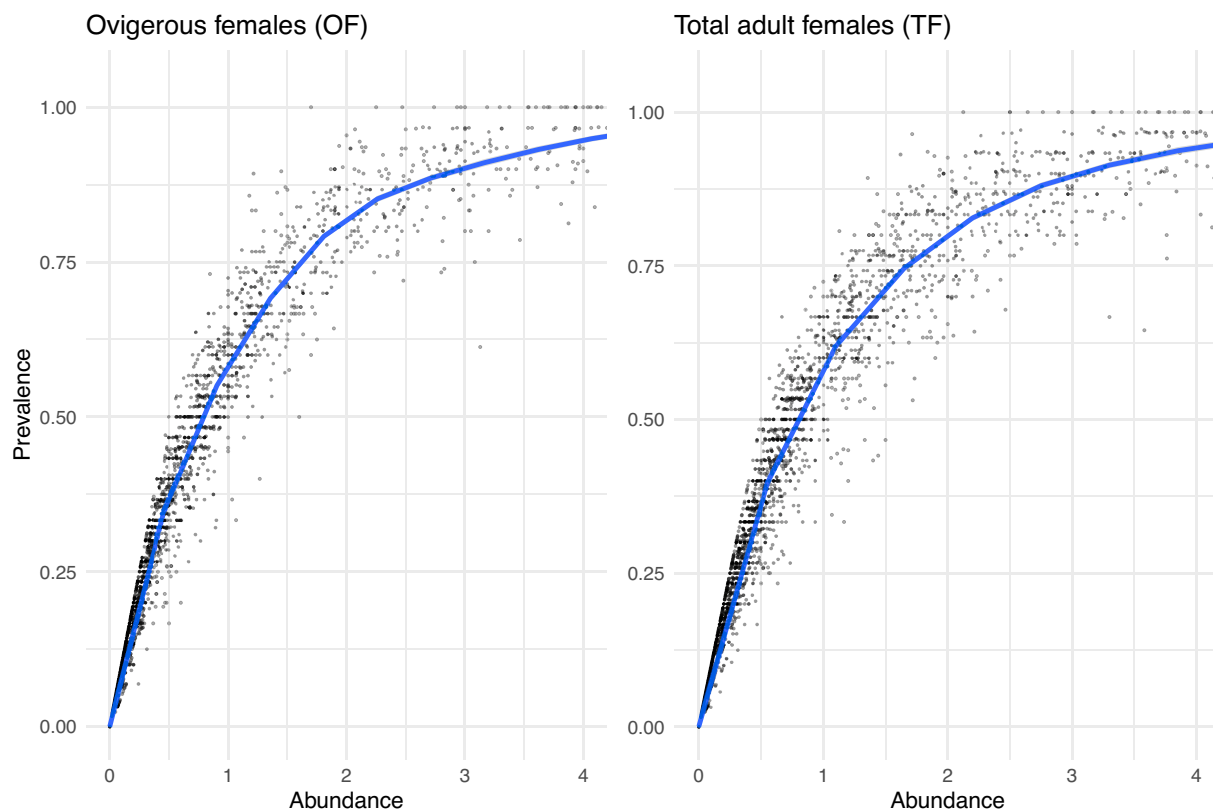


Fig. 2. Scatterplot of the relationship between mean abundance and prevalence of ovigerous females (OF, left-hand graph) and total adult females (TF, right-hand graph) per fish at the pen level. The data are fitted with a smoother line with 'gam' function (blue). The data were recorded under the NSLMP. For better visualisation all mean abundance values of both OF and TF higher than 4 were removed ($n = 6098$; 3.6 % OF; 5.4 % of TF means per fish at the pen level were removed). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

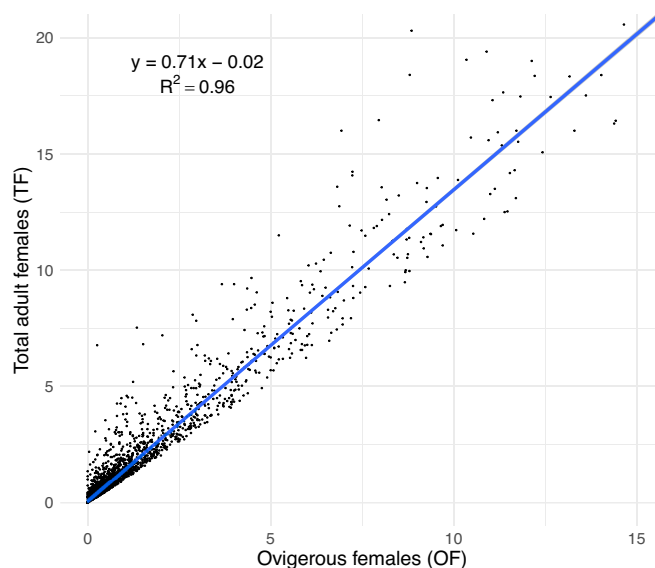


Fig. 3. Scatterplot of the relationship between the mean abundance of ovigerous females (OF) and total adult females (TF) per fish at the pen level. The data are fitted with regression line (blue) with the equation and correlation coefficient (R^2) displayed in the left-hand corner. The 95 % confidence interval is highlighted in dark grey, mostly hidden by the regression line. The data were recorded under the NSLMP. For better visualisation all mean values higher than 15 for OF and higher than 20 for TF were omitted ($n = 6098$, 0.2 % OF, 0.3 % TF means per fish at the pen level were removed). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

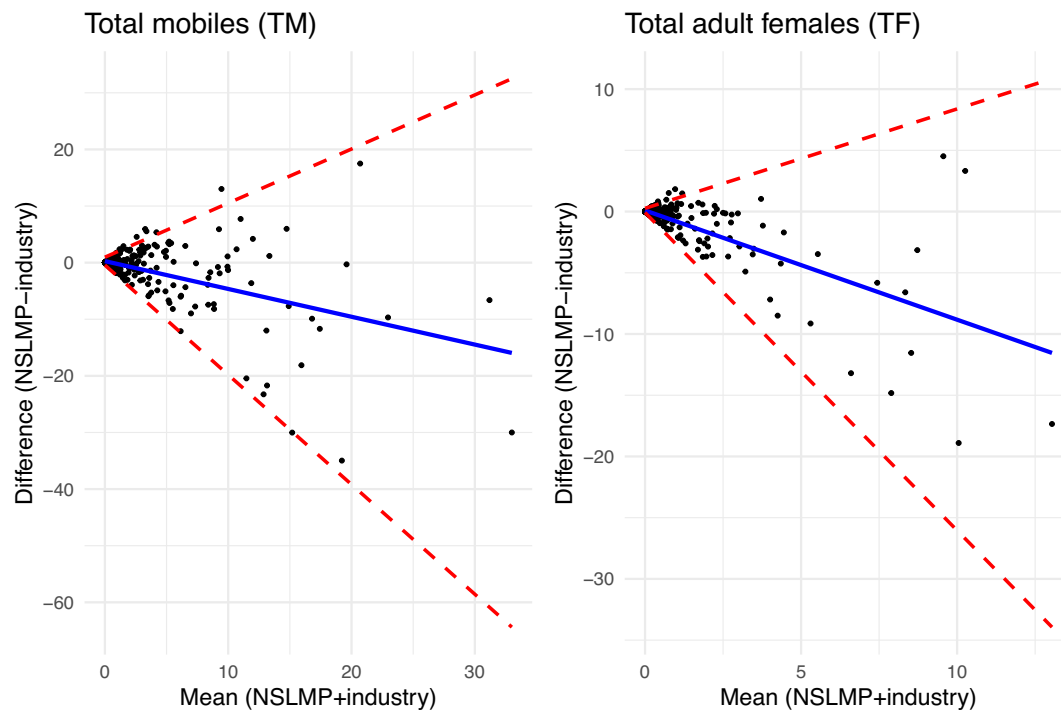


Fig. 4. Bland-Altman plots of mean sea lice abundance per fish at the pen level (separately for total adult females (TF) and total mobiles (TM)) recorded under the NSLMP and by an industry producer. The 95 % confidence interval is highlighted by the red dotted line. For better visualisation all mean values higher than 25 were removed ($n = 876$, 0.9 % TM means per fish at the pen level were removed). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

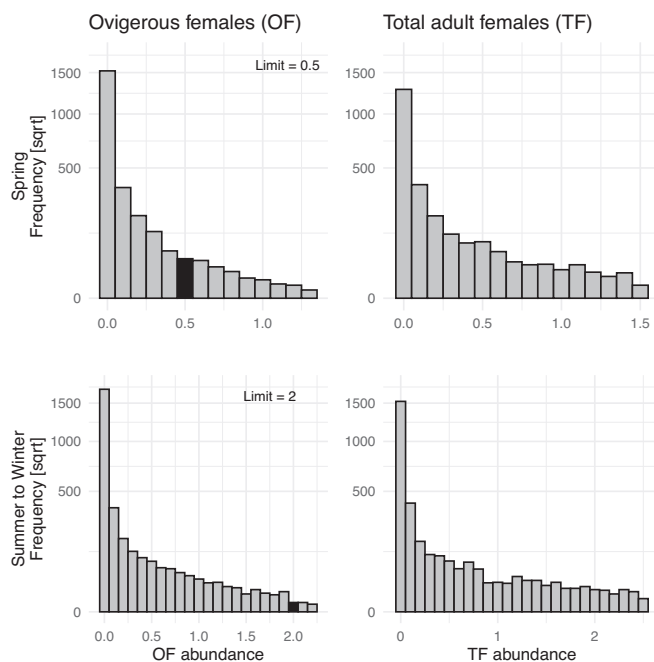


Fig. 5. Histograms of mean abundance of ovigerous females (OF, left-hand panels) and total adult females (TF, right-hand panels) per fish at the pen level divided by seasons. The first row corresponds to spring months (March, April, May) with a treatment trigger level of 0.5 OF per fish at the pen level on average. The second row corresponds to the summer to winter months with a trigger level of two OF per fish at the pen level on average. Trigger levels are highlighted in black. Data were recorded under the NSLMP. For better visualisation all TF mean values higher than 1.5 in spring ($n = 2384$, 7.7 % TF means per fish at the pen level were removed) and three in summer to winter ($n = 3155$, 10.1 % TF means per fish at the pen level were removed) were omitted.

interval [0.07, 0.22] for total adult females and -0.15 with 95 % Confidential interval [0.07, 0.22] for total mobiles.

The Bland-Altman plots (Fig. 4) showed that the reported values by NSLMP and industry producer tend to agree at low abundance levels. However, there is an increase in variance with the increase of detected abundance. And in case of higher sea lice abundance, NSLMP tend to report lower abundance than the industry producer. The Wilcoxon matched pairs test confirmed the results of Bland-Altman plot, detecting no significant differences between the reported abundance of total mobiles between the NSLMP and Industry producer ($V = 76,772$, $p = 0.07$). However, significant differences were detected for the reported values of total adult females ($V = 51,616$, $p \leq 0.01$). The estimated median difference was 0.06 (95 % CI: 0.03 to 0.09).

3.4. Influence of treatment trigger levels

The reported sea lice abundance of ovigerous females and total adult females in the data recorded under the NSLMP were explored separately for the spring period (March – May) and summer to winter (Fig. 5). The reported values of both ovigerous females and total adult females are evenly distributed and show no distinctive decrease of counts around the treatment trigger levels (spring: 0.5, summer to winter: two ovigerous females). The reported abundance of total adult females was also compared between the NSLMP and industry producer's data (Fig. 6). The reported values for both datasets showed no distinctive decrease of counts around the industry producer's treatment trigger level (0.2 of total adult females).

4. Discussion

Sea lice monitoring is considered a fundamental part of management strategies used on salmon farms (Treasurer and Pope, 2000; Jeong et al., 2023). Monitoring enables estimation of sea lice population levels providing necessary data for effective and timely treatment (Westcott et al., 2004). This study evaluated two metrics of sea lice

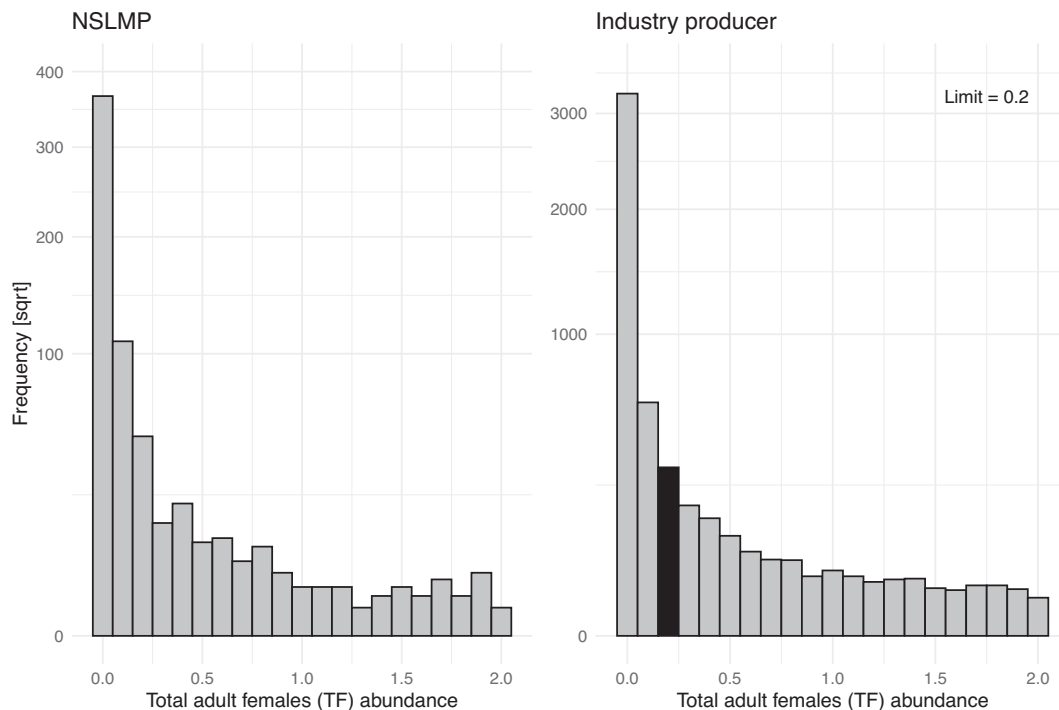


Fig. 6. Histograms of mean abundance of total adult females (TF) per fish at the pen level recorded under the NSLMP (left-hand panel) and by industry producer (right-hand panel). The treatment trigger level used by industry throughout the whole year is the mean value of 0.2 TF per fish at the pen level and is highlighted in black. For better visualisation all pens with a higher mean value than two TF per fish were removed (NSLMP: $n = 891$, 4.2 % TF values were removed; Industry producer: $n = 7437$, 9 % TF values were removed).

infection—abundance and prevalence—to determine their suitability within monitoring programmes. It also examined the potential differences in the abundance of different sea lice life stages used for treatment trigger levels and explored the relationship between ovigerous females and total adult females in Irish datasets. Furthermore, the study highlighted discrepancies in reported sea lice abundance between independent monitoring recorded as part of the National Sea lice Monitoring Programme (NSLMP) and one industry producer's assessments. Finally, the study investigated how treatment trigger levels influence the accuracy of reported sea lice counts by visually examining these data.

4.1. Prevalence as a metric of sea lice infection

Analysis of sea lice data recorded as part of NSLMP revealed a linear relationship between mean sea lice abundance and prevalence up to ~40 %. Beyond this threshold, abundance increased more rapidly, consistent with patterns expected under a Poisson and negative binomial distribution, as described in previous studies (Patanasatienkul et al., 2013; Jeong and Revie, 2020). While this initial linearity aligns with previous studies; those studies reported a linearity phase extending to higher prevalence levels (up to ~70 %) (Heuch et al., 2011; Patanasatienkul et al., 2013; Jeong and Revie, 2020). It should be noted that the estimated threshold for the breakdown of linearity is approximate and may carry some uncertainty. The observed discrepancy between regions could suggest greater sea lice aggregation on fish in Ireland compared to other areas, although further analysis is required to confirm this. Previous research has linked parasite aggregation to stocking density (Krasnov et al., 2007; Samsing et al., 2014); however, due to the absence of data on fish population size, cross-country comparisons of sea lice aggregation were not possible.

The observed linearity between sea lice abundance and prevalence suggests that monitoring programs on salmon farms could potentially be simplified by shifting from abundance-based to prevalence-based assessments. Under this approach, only the presence or absence of sea lice on each fish would be recorded, rather than counts of all life stages. Such

a shift would likely result in minimal information loss, particularly when overdispersion is low, because abundance can be reliably estimated from prevalence (Jeong and Revie, 2020). Although prevalence measures may introduce slight bias by excluding detached lice, this limitation is consistent with current industry practice, as detached lice are not included in abundance estimates. Apart from that, prevalence-based monitoring offers clear advantages: it is more resource-efficient, reduces the effort required to detect sea lice — especially smaller life stages, which are challenging to count accurately (Treasurer and Pope, 2000; Baillie et al., 2009) — and substantially decreases inspection time. Shorter handling times also reduce stress on fish by limiting the period they remain out of water (Baillie et al., 2009). However, empirical validation of the efficiency and welfare benefits of this approach remains lacking and should be a focus of future research.

However, issues can arise in the case of high infection levels. Not only is it not possible to estimate abundance from prevalence, because the relationship is no longer linear (prevalence higher than ~40 %) but recording only prevalence also leads to a loss of information on sea lice population size. High infection levels often reach prevalences of 100 % but abundance levels can differ substantially. Therefore, relying on prevalence alone is not possible for a sea lice monitoring programme. On the other hand, publishing information on prevalence in addition to abundance gains the most information and is not more time consuming than recording only abundance. This approach is especially beneficial for the determination of the aggregation of sea lice within pens, which was recognised as an issue on salmon farms in previous studies (Baillie et al., 2009; Jeong and Revie, 2020).

For that reason, recording only prevalence could be implemented in cases of low infection levels (prevalence < ~40 %) or as a resource-efficient approach where required, for example to shorten the time of inspections. Prevalence measures could also be used as a time-efficient approach to increasing the number of sampled pens where, for instance, the same number of pens could be monitored for sea lice abundance, but all pens could be monitored for sea lice prevalence.

4.2. Relationship between ovigerous females and total adult females

Although most countries use the counts of total adult females for their treatment trigger levels during sea lice monitoring on salmon farms, Ireland focuses on the counts of ovigerous females (Guarracino et al., 2018; Vormedal and Larsen, 2021; D'Arcy et al., 2024). The current treatment trigger levels used in Ireland are a mean of 0.5 and 2 ovigerous females per fish at the pen level in spring and summer to winter, respectively. Here we compared data for ovigerous females and total adult females and were able to estimate the equivalent values for total adult females which resulted in a mean of 0.7 and 2.8 of total adult females per fish at the pen level in spring and summer to winter, respectively. Comparing these approximated treatment trigger levels of total adult females to other European producers shows that the treatment trigger level in both Norway (0.2 spring, 0.5 summer to winter) and in the Faroe Islands (0.5 spring, 1 summer to winter) are lower (Guarracino et al., 2018; Vormedal and Larsen, 2021). Similar levels are used in Scotland (0.5 spring, 1 summer to winter), with monitoring frequency increased if abundance exceeds 2 total adult females (CoGP, 2015; Vormedal, 2024). Since higher salmon densities are believed to enhance sea lice transmission (Krkošek, 2010), countries with greater production volumes and therefore larger fish populations may require stricter treatment thresholds to maintain control. This likely explains why Norway, Scotland and the Faroe Islands apply lower thresholds despite producing far more salmon than Ireland: 1.2 million tonnes, 151,000 t and 80,000 t annually, respectively, compared with only 9300 t in Ireland (Food Business, 2021; Dennis et al., 2024; Scottish Government, 2024).

Standardizing sea lice reporting and treatment trigger levels based on a single life stage across all Atlantic salmon-producing countries would enable more consistent global comparisons of infection levels. Currently, Ireland is the only European country that reports ovigerous female counts. Shifting to total adult females would align Ireland's data with that of other major producers, such as Norway, the Faroe Islands, and Scotland. However, such a change would disrupt Ireland's long-term time series, which represents a uniquely valuable public dataset that has been maintained for decades, although counts of total adult females are collected and available if required.

Different life stages may also be monitored for different reasons: counts of total mobile lice can be used to assess fish welfare impacts, whereas ovigerous females provide an indication of egg production and the potential for onward transmission (Treasurer and Pope, 2000; Rabe et al., 2024; Vormedal, 2024). Reporting total adult females instead of ovigerous females has been recommended, as it reduces uncertainty by providing higher counts (since ovigerous females are a subset of total adults), minimizes potential underreporting due to egg string loss during inspections and might potentially prevent the production of larval stages (Rabe et al., 2024). Applying Ireland's current treatment trigger levels to total adult females (0.5 in spring, 2 from summer to winter) rather than ovigerous females would result in stricter control measures. These thresholds would correspond to mean values of 0.3 and 1.4 ovigerous females per fish at the pen level in spring and summer to winter, respectively.

4.3. A comparison of reported sea lice counts

Sea lice counts reported under the NSLMP and by the industry producer generally align at low abundance levels. However, as abundance increases, the variation between the two datasets becomes more pronounced. At higher abundance levels, NSLMP counts tend to be lower than those reported by industry producers. However, significant differences between the two monitoring programs were observed only for total adult females.

Although previous studies have highlighted under-reporting of sea lice infection levels by the salmon industry producers (Elmoslemany et al., 2013; Godwin et al., 2021), the results of this study revealed

higher reported abundances of total adult females from an industry producer compared to independent monitoring recorded as part of the NSLMP. Differences in methodologies between the two monitoring approaches may account for some of the variation between datasets, though they are unlikely to explain all the differences. The number of fish sampled at the pen level varied between the two methods (30 for NSLMP and 20 for industry producer), as did the techniques used for sea lice counting (NSLMP: collection of sea lice; industry producer: visual examination). Sampling a higher number of fish provides a more robust population estimate and reduces the influence of fluctuations caused by sea lice aggregation or non-random fish selection (Andrade, 2020; Mes et al., 2024; Rabe et al., 2024). Additionally, visual assessments of sea lice levels, as used by the industry, may introduce errors into the reported counts (Rabe et al., 2024).

Although 45 % of the industry data were collected on the same date as the NSLMP inspections, the remaining data were sampled either before or after the NSLMP inspections (however within 7 days). When the sampling dates were further apart, it is possible that treatments were applied prior to an NSLMP inspection, potentially influencing the overall sea lice population. However, only a weak correlation was observed between the time difference in sampling dates and differences in reported sea lice counts. Additionally, when both the NSLMP and industry conducted inspections on the same date, sea lice counts were randomly chosen from a different pen. As a result, some detected differences in reported counts could be attributed to variations in sea lice levels between pens (Treasurer and Pope, 2000; Marques et al., 2018). Furthermore, there is no incentive for underreporting for either the industry producer that uses monitoring for internal management and for a third-party monitoring conducted by the Marine Institute as part of the NSLMP.

4.4. Influence of treatment trigger levels

The call for more stringent control measures against sea lice on Atlantic salmon farms can lead to very low and ambitious treatment trigger levels (Gislason, 2018; Vollset et al., 2018), which could result in the under-reporting of sea lice infection levels (Jeong et al., 2023; Telford, 2025). The treatment trigger levels applied on Irish salmon farms are not as low as in Norway, Scotland or the Faroe Islands (Guarracino et al., 2018; Vormedal and Larsen, 2021; Vormedal, 2024). In addition, the independent producers have their own treatment trigger levels, for example Mowi Ireland uses the treatment trigger level of a mean of 0.2 total adult female lice per fish at the pen level throughout the year.

The exploration of the sea lice counts reported by the NSLMP and from the industry producer showed no inconsistencies between the life stages and the number of reported sea lice counts around the treatment trigger levels. Therefore, it appears that the currently used treatment trigger levels do not have a negative impact on the accuracy of reported values.

5. Conclusions

This study analysed an extensive dataset of sea lice abundance from salmon farms in Ireland to examine the relationship between abundance and prevalence, compare life-stage reporting across countries, and evaluate differences between industry-reported counts (Mowi Ireland) and independent monitoring (NSLMP, Marine Institute). Our findings highlight the benefits of recording prevalence alongside abundance, the potential advantages of standardizing life-stage reporting, and the implications for treatment trigger levels. Minor discrepancies were observed between industry and NSLMP reports at low abundance, with greater variability at higher sea lice levels. These results underline the importance of robust, standardized monitoring strategies to accurately assess sea lice populations on salmon farms. Monitoring practices are continuously being refined, and in recent years, advanced camera-based technologies have been introduced (Pettersen et al., 2019; Stien et al., 2020; Rabe et al., 2024). This approach offers continuous estimation of

sea lice population levels in addition to minimal effect on welfare of fish due to the lack of manual handling and anaesthetics (Rabe et al., 2024). Looking ahead, integrating camera-based systems into sea lice monitoring could complement existing approaches, enabling faster, more accurate, and welfare-conscious assessment of sea lice populations on salmon farms.

CRediT authorship contribution statement

Bela Klimesova: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Conceptualization. **Katie O'Dwyer:** Writing – review & editing, Validation, Methodology. **Jack D'Arcy:** Writing – review & editing, Writing – original draft, Data curation, Conceptualization. **Anita Talbot:** Writing – review & editing, Project administration, Funding acquisition. **Olga Lyashevskaya:** Writing – review & editing, Validation. **Hamish Rodger:** Writing – review & editing. **Catherine McManus:** Writing – review & editing, Data curation. **Neil M. Ruane:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Conceptualization.

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Declaration of competing interest

Catherine McManus is an employee of Mowi Ireland, a company from which some of the data used in this study originates. All analyses and interpretations presented in this paper were conducted independently and without influence from Mowi Ireland. The remaining authors declare no conflicts of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://github.com/Klimesobe/Metrics_Sea_Lice_Infection.git.

Data availability

The data and R code used in this study are available at <https://doi.org/10.6084/m9.figshare.30466883> and https://github.com/Klimesobe/Metrics_Sea_Lice_Infection.git.

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